

VYOMA KOTAK

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PROFESSIONAL SUMMARY

Data scientist and engineer with over four years of professional experience turning large, complex datasets into reliable, decision ready insight. I build complete data science solutions end to end, from engineering distributed pipelines over tens of millions of records to training, evaluating, and explaining models that hold up under real world scrutiny. I am fluent in Python and SQL and work across supervised learning, NLP, and generative AI, using tools such as scikit learn, XGBoost, PyTorch, Spark, and FastAPI. My background in production systems gives me a strong instinct for reliability and clean engineering, so the solutions I build are not just accurate but dependable. Above all, I translate technical findings into clear insight that both technical and business audiences can act on, and I thrive on solving hard problems in fast moving, high impact environments.

TECHNICAL SKILLS

- **Programming:** Python (Pandas, NumPy, PySpark, scikit-learn, XGBoost), SQL, Java
- **Data Engineering:** ETL Pipelines, Apache Airflow, Data Modelling, Structured & Unstructured Data, Distributed Computing
- **Machine Learning:** Supervised & Unsupervised Learning, Classification, NLP, Feature Engineering, Model Evaluation, Model Explainability (SHAP)
- **Deep Learning & GenAI:** PyTorch, TensorFlow/Keras, Neural Networks (CNNs, RNN/LSTM), LLMs, Generative AI
- **Big Data & Streaming:** Apache Spark, Spark Structured Streaming, Kafka
- **Cloud & Deployment:** AWS (S3, EC2, SageMaker), Google Cloud Platform (BigQuery), Docker, FastAPI, Git, CI/CD, REST APIs
- **Databases:** Advanced SQL, Oracle, MySQL
- **Visualisation:** Power BI, Tableau, Advanced Excel
- **Core Strengths:** Production ML Pipelines, Stakeholder Communication, Problem Solving, Reproducible Analysis

PROFESSIONAL EXPERIENCE

Analytics Leap | Coventry, UK

Data Analyst

March 2026 – Present

- Analysed UK data analytics job market trends, employer requirements, and skill gaps to support data driven curriculum decisions and align content with industry needs.
- Designed and developed practical analytics content covering Python, SQL, Power BI, Tableau, and Excel, helping learners build job ready data skills using real world datasets.
- Created 50+ hands on exercises, case studies, and dashboard projects on structured datasets to improve engagement and practical application.
- Reviewed and optimised existing materials against industry best practices, improving accuracy and reducing outdated technical references by 30%.
- Translated complex analytical concepts into clear workflows, improving learning accessibility for beginner and intermediate users.

Tata Consultancy Services | India

System Engineer – Python Developer

March 2024 – January 2026

- Built and maintained Python web applications and REST APIs using Django and Flask, delivering features supporting internal business processes and client requirements across regular release cycles.
- Designed OTP based authentication and user verification APIs that strengthened account security and streamlined login and password reset flows.
- Tuned slow API endpoints by reworking database queries and application logic, cutting average response times by roughly 30 to 40%.
- Automated repetitive data processing and reporting tasks with Python and SQL, saving the support team an estimated 8 to 10 hours of manual work each week.
- Collaborated with QA, frontend, and business colleagues in an Agile team, troubleshooting production issues and keeping releases on schedule.
- Added unit tests and debugging scripts that improved code reliability and reduced post release issues.

- Took hands on ownership of SAP ECC, NetWeaver (PI/PO), and dual stack BW/BPC systems for global enterprise clients, keeping them stable and available around the clock.
- Carried out system refresh and homogeneous copy cycles to keep non production environments aligned with production, reducing deployment risk.
- Applied critical SAP Notes, kernel upgrades, and support packages through SUM, closing security gaps and improving platform stability.
- Investigated and resolved database locks, memory bottlenecks, and long running jobs, improving performance and reducing downtime.
- Set up proactive monitoring through SAP Solution Manager (EWA, dashboards) and maintained DR and backup procedures.

PROJECTS

Real-Time Fraud Detection Platform

Python, Kafka, Spark Structured Streaming, FastAPI, Docker

github.com/VyomaKotak/Real-Time-Fraud-Detection-Platform

- Designed an end to end streaming platform to score 6.3M mobile money transactions in real time, detecting fraud within milliseconds against a 0.13% fraud base rate.
- Trained a Random Forest classifier reaching PR-AUC 0.998, catching 1,638 of 1,643 frauds at just 41 false positives, with domain driven balance error features contributing over a third of the model's signal.
- Engineered a single shared feature module across training, the FastAPI service, and the Spark scorer to eliminate train/serve skew, containerising the full pipeline with Docker and CI (26 passing tests).

Large-Scale Amazon Review Sentiment Classification

Python, PySpark, Spark MLlib, SHAP, Tableau

github.com/VyomaKotak/Amazon-Reviews-Sentiment-Classification

- Built a distributed PySpark pipeline over 43.9M Amazon Electronics reviews (21 GB), turning unread customer feedback into an automated sentiment signal at scale.
- Engineered a six stage ML pipeline (Tokenizer, StopWords, TF-IDF, Scaler) and benchmarked four classifiers, with Logistic Regression winning at F1 0.851 and AUC 0.915.
- Optimised distributed performance through caching, repartitioning, and skew analysis, and explained the model with SHAP alongside four Tableau dashboards.

EDUCATION

MSc Data Science

Coventry University, UK

January 2026 – Present

BE Electronics and Communication | 8.61 CGPA

Gujarat Technological University, India

August 2017 – May 2021

ADDITIONAL ACHIEVEMENTS

- AI Fluency: Framework & Foundations – Anthropic (May 2026)
- SAP Certified Specialist – SAP S/4 HANA Conversion and System Upgrade
- BCG X Data Science Job Simulation (Forage) – PowerCo churn analysis
- Elevate Wings1 'Hi-Talent' – TCS: Outstanding performance recognition
- Life Skills Mentor – Kotak Education Foundation: communication and employability skills
- On the Spot (Team) Award – TCS: Excellence under pressure
- Service Commitment Award – TCS: 3 years of impactful contribution